

Physics-Aware Multimodal Urban Heat Mapping with Open Web Imagery and Mobility Data

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Abstract

Extreme urban heat is intensifying worldwide and often falls hardest on vulnerable communities, posing growing challenges for climate adaptation and Sustainable Development Goal 11. Fine-grained land surface temperature (LST) estimates are essential for identifying local heat risks, yet most operational approaches still rely on satellite products alone, which are constrained by cloud cover, revisit cycles, and limited sensitivity to human-scale morphology and activity. Meanwhile, web-based resources such as online imagery and mobility data offer rich but underused signals for scalable heat-risk monitoring. We present **AESPA**, a physics-aware multimodal framework for tract-level urban LST estimation that combines satellite imagery, street-view panoramas, mobility-derived activity profiles, and interpretable physical proxies. AESPA trains a teacher model that jointly leverages imagery and mobility, then distills its predictions and representations into an imagery-only student, enabling deployment in data-poor cities. Physics- and proxy-guided losses encourage consistency with basic urban-climate relationships and yield attributions linked to vegetation, impervious surfaces, shading, and surface reflectance. We evaluate AESPA across eight major U.S. metropolitan areas under within-city and cross-city transfer protocols: AESPA reduces mean absolute error by about 32% and increases Pearson correlation between predicted and observed tract-level LST by 0.15 compared with the strongest satellite-based baseline, and improves transfer correlations by roughly 0.05-0.10. Its proxy attributions recover physically coherent gradients associated with neighborhood-level heat-exposure inequality, illustrating how web-based imagery and mobility can support transparent, deployable urban heat-risk monitoring in practice¹.

CCS Concepts

• **Computing methodologies** → **Computer vision**; • **Human-centered computing** → *Collaborative and social computing*; • **Information systems** → **Spatial-temporal systems**.

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¹The code and data are available at <https://github.com/tsinghua-fib-lab/AESPA>.



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Keywords

Urban heat estimation, multimodal learning, human mobility, knowledge distillation, climate equity

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1 Introduction

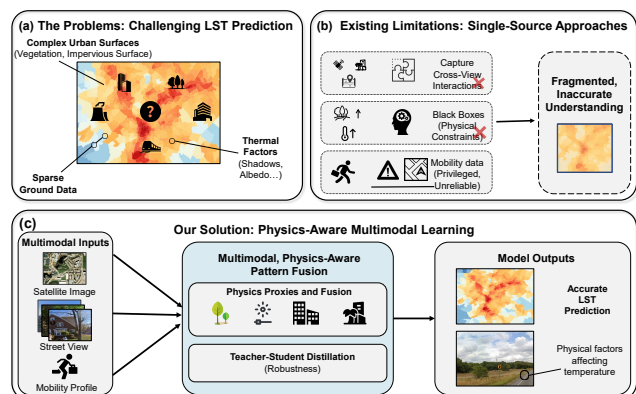


Figure 1: Overview of the motivation of our work. Our model fills the gap of multimodal, physics-aware learning on the abundant thermal information across multi-view urban data from web platform to accurately predict land surface temperature.

Extreme heat is intensifying under climate change, with cities emerging as critical hotspots due to the urban heat island (UHI) effect. Elevated urban temperatures exacerbate heat-related morbidity and mortality, strain energy and water systems, and disproportionately burden low-income, elderly, and marginalized communities [1, 14, 21, 35, 54]. Recent work further shows that heat *exposure* depends not only on where people live but also on where they travel during the day, as daily mobility concentrates people in persistent “heat traps” [12, 23]. Delivering the Sustainable Development Goal on sustainable cities and communities (SDG 11) [29] therefore requires reliable, fine-grained information on how heat is distributed across urban neighborhoods and how it interacts with the built environment and human activity. Yet city agencies and

community organizations often lack tract-level temperature maps that are timely, interpretable, and scalable, despite the growing availability of satellite imagery, street-view image, and open urban data from *web platforms* [40].

A primary line of work in urban heat mapping estimates land surface temperature (LST) or near-surface air temperature from satellite observations. Early methods use statistical downscaling and physical retrieval algorithms that combine thermal infrared bands with vegetation and built-up indices in regression or split-window frameworks [8, 39]. More recent deep models leverage multi-source satellite products and auxiliary geophysical variables to improve prediction accuracy at finer spatial resolutions: Urban-Heat [36] predicts fine-grained urban air temperature from multi-resolution imagery with physically inspired objectives, and Deep-UHI [55] introduces a context-aware thermodynamic framework for urban heat island forecasting, while other work couples split-window algorithms with neural networks or interpretable downscaling pipelines for LST retrieval [5, 10]. These approaches demonstrate the power of satellite imagery, but remain largely satellite-centric: they are constrained by coarse thermal resolution and two-dimensional surface proxies, with limited representation of street-level morphology, shading, and human behaviour.

At the same time, *Web platforms* now provide rich cross-view data about cities. Street-view imagery has been used to derive green view, sky view and façade indices and relate them to neighborhood-scale LST and microclimate [38, 42, 53], while multimodal and vision-language models leverage satellite and street-level images to infer socioeconomic status, segregation, health outcomes and land use [17–20, 22, 30, 50, 52]. In parallel, mobility-based studies show that daily travel patterns concentrate people in persistent “heat traps” and strongly shape heat exposure beyond residential locations [12, 23]. Together, these developments suggest that satellite imagery, street-level morphology and human activity patterns jointly encode the fine-grained urban thermodynamics that matter for climate adaptation.

However, existing methods only partially exploit this multimodal context, leading to three concrete challenges. First, *multimodal context is fragmented*. Most temperature models operate on satellite imagery alone or add a few handcrafted street-view indices, without coherently integrating land cover, street-level form and mobility. Second, prevailing deep regressors are largely *physics-agnostic black boxes*. Without explicit ties to basic urban-climate mechanisms, they can violate simple thermal intuition (e.g., predicting higher temperature when vegetation increases), limiting their usefulness for planning [41]. Third, *rich auxiliary signals such as mobility are unevenly available*. Mobility profiles are informative for exposure but often sparse, sensitive or missing, calling for models that can use them at training time without requiring them at deployment. These gaps motivate the need for a physically grounded, multimodal framework that can turn web-sourced imagery and mobility into reliable, tract-level urban temperature maps.

In this paper, we propose **AESPA** (Aligned Environmental Sensing with Physics-aware Attribution), a multimodal framework for fine-grained urban temperature estimation at the census-tract level. AESPA encodes satellite patches and sets of street-view panoramas for each tract using strong vision backbones, and fuses them into a tract-level representation via a lightweight transformer. From the

same street-view pixels, it derives five physically motivated proxies that summarize tract-level street context: vegetation, tree canopy, impervious surface, apparent albedo, and shadow. These proxies serve a dual role: they are predicted by dedicated heads as auxiliary tasks, and they drive simple physics-aware regularizers that enforce monotone relationships between proxies and predicted temperature (e.g., more vegetation and canopy should not systematically raise predicted temperature, while higher imperviousness should not systematically cool it). To leverage richer signals where available without compromising deployability, AESPA trains a mobility-aware teacher model and distills its predictions and fused representations into an imagery-only student, treating mobility as *privileged information* for cross-city generalization. Figure 1 summarizes the motivation of the AESPA model and how physical proxies align cross-view imagery with urban climatic mechanisms.

We evaluate AESPA across eight metropolitan statistical areas (MSAs) in the United States, using summer daytime land surface temperature at the census-tract level as the prediction target. We benchmark AESPA against strong satellite-only and proxy-based baselines, as well as ablated variants without certain modalities (satellite, street view, mobility) and without physics-aware losses, proxy heads, or teacher–student distillation. Across MSAs, AESPA achieves the lowest or second-lowest error in almost all cities, reducing average MAE from 1.95 °C for the best baseline to 1.33 °C and increasing the mean tract-level Pearson correlation from 0.61 to 0.76 in within-MSA experiments. In cross-MSA transfer, it further improves correlation by about 0.05–0.10 over imagery-only baselines, indicating stronger generalization to unseen cities. A tract-level case study in Dallas shows that AESPA more faithfully reproduces socioeconomic gradients in heat exposure than ResNet, supporting analysis of disparities across neighborhoods with different racial and poverty compositions. In summary, our main contributions are threefold:

- **Multimodal framework for tract-level temperature mapping.** AESPA integrates satellite imagery and street-view panoramas (plus mobility profiles during training) via vision encoders and cross-feature fusion to predict summer daytime temperature at the census-tract level across eight U.S. metropolitan areas.
- **Street-view physical proxies and physics-aware regularization.** From street-view pixels, AESPA derives five interpretable proxies (vegetation, canopy, imperviousness, apparent albedo, shadow) and uses them as auxiliary heads and sign-constrained regularizers, improving robustness and interpretability without requiring a full thermodynamic simulator.
- **Mobility-aware teacher–student distillation for imagery-only deployment.** AESPA treats mobility as privileged training information in a teacher model and distills its predictions and representations into an imagery-only student, enabling better cross-MSA generalization while requiring only widely available satellite and street-view imagery at deployment time.

2 Related Works

2.1 Satellite-Based Temperature Estimation

Accurate measurement of urban temperature is crucial for understanding the heat island effect and promoting sustainable cities (SDG 11) [29]. A primary thrust in this area involves estimating

land surface temperature and air temperature from satellite imagery. Early approaches relied on statistical downscaling and physical retrieval algorithms, combining thermal infrared data with vegetation or urban indices within traditional regression frameworks [8, 39]. While computationally efficient, these methods are inherently limited in capturing the complex, fine-grained thermal variations driven by street-level urban morphology. The advent of deep learning has enabled more sophisticated fusion of multi-source satellite products. Contemporary work explores the use of multi-resolution imagery to predict urban air temperature through physically inspired loss functions [36], while other approaches integrate thermodynamic context for forecasting urban heat island dynamics [55]. Further innovations include hybrid models that couple physical radiative transfer equations with neural networks for robust temperature retrieval [5], as well as interpretable downscaling pipelines designed for urban environments [10].

These advances have substantially improved satellite-based temperature mapping, but they remain predominantly satellite-centric and do not exploit street-level views or human activity. In contrast, we treat satellite imagery as one component within a multimodal framework, combining it with street-view cues and simple physical constraints to model intra-urban temperature at finer scale.

2.2 Street-View Imagery and Urban Microclimate

Street-view imagery provides a ground-level perspective on urban form, capturing features such as vegetation density and street canyon geometry that directly influence microclimate. Initial research focused on deriving quantitative indices—such as green view or sky view factor—and correlating them with neighborhood-scale temperature patterns using spatial statistical models [53]. Subsequent studies have integrated these street-level features with traditional land cover data to improve temperature modeling, often within single-city contexts [38, 42]. Beyond microclimate, the computer vision community has leveraged joint analysis of satellite and street-view imagery for urban inference tasks, including socioeconomic assessment [22, 52], health prediction [30], and land use classification [17, 50].

This body of work highlights the value of street-view imagery, but most microclimate applications rely on a small number of pre-defined indices used as auxiliary covariates. Our approach instead derives physically grounded proxies directly from street-view pixels and uses them both as conditioning features and auxiliary learning targets, allowing street-level signals to regularize cross-modal temperature prediction in a more systematic way.

2.3 Physics-Constrained Environmental Machine Learning

The integration of physical knowledge into machine learning has gained increasing attention as a means to improve the robustness and generalization of environmental and urban sensing systems. In the context of land surface temperature and urban heat estimation, recent frameworks have sought to embed physical principles—such as surface energy-balance constraints or radiative transfer equations—directly into learning-based models. Examples include coupling split-window physical algorithms with neural networks for improved

LST retrieval [5], incorporating thermodynamic regularizations into downscaling and forecasting architectures [4, 27], and using physics-inspired objectives to guide model training. Beyond urban heat, similar methodology has been applied to tasks such as human activity prediction [34, 49] and high-resolution carbon emission estimation [47, 48] from open data, underscoring the broad applicability of physics-aware learning in sustainability contexts. Surveys of this emerging field further emphasize the value of embedding interpretable physical components into data-driven urban modeling pipelines [41].

While these approaches demonstrate improved consistency with domain knowledge, many depend on detailed physical forcings, high-resolution 3D geometry, or task-specific simulators and are typically designed for gridded remote-sensing inputs. We instead adopt a lightweight physics-aware design that encodes well-established monotonic relationships (e.g., between vegetation and temperature), together with optional spatial smoothness and day–night ordering, as soft constraints within a multimodal encoder. This aims to preserve physical plausibility and interpretability while remaining compatible with web-based satellite and street-view imagery for scalable urban temperature estimation.

3 Methods

3.1 Problem Definition

We aim to estimate fine-grained urban temperature at the level of census tracts across multiple U.S. metropolitan areas. Let $\mathcal{T}$ denote the set of all tracts in our study, and index each tract by $i \in \mathcal{T}$. For each tract i we obtain three types of inputs that characterize its urban context: (1) a high-resolution satellite image $I_i^{\text{sat}} \in \mathbb{R}^{H \times W \times 3}$ centered on the tract; (2) a set of K street-view images $I_i^{\text{sv}} = \{I_{i,1}^{\text{sv}}, \dots, I_{i,K}^{\text{sv}}\}$ sampled along roads within the tract, capturing human-scale morphology such as building facades, trees, and shading structures; (3) a 168-dimensional mobility profile $\mathbf{m}_i \in \mathbb{R}^{168}$ that summarizes hourly human activity over summer weeks.

For each tract i , we observe a scalar target $y_i \in \mathbb{R}$, defined as the average summer daytime temperature over the study period. The learning task is to estimate a function $f_\theta : (I_i^{\text{sat}}, I_i^{\text{sv}}, \mathbf{m}_i) \mapsto \hat{y}_i$ parameterized by θ , such that the prediction $\hat{y}_i$ approximates the observed temperature y_i .

3.2 Model Overview

We propose AESPA, a multimodal framework for tract-level urban temperature prediction. Building on the inputs defined in Section 3.1, it first encodes each modality with a dedicated encoder to obtain compact satellite, street-view, and mobility representations. These representations are merged by a cross-feature fusion block that models their interactions and produces a single tract embedding, from which the model regresses daytime temperature. To improve robustness and interpretability, we attach lightweight latent heads that recover physically meaningful factors from the same embedding and use them only during training to regularize the temperature mapping. Finally, we adopt a teacher–student design: a mobility-aware teacher is trained with all modalities, and an imagery-only student is distilled from it. The student branch, which relies only on imagery, is used at inference time. Figure 2 illustrate the overall model.

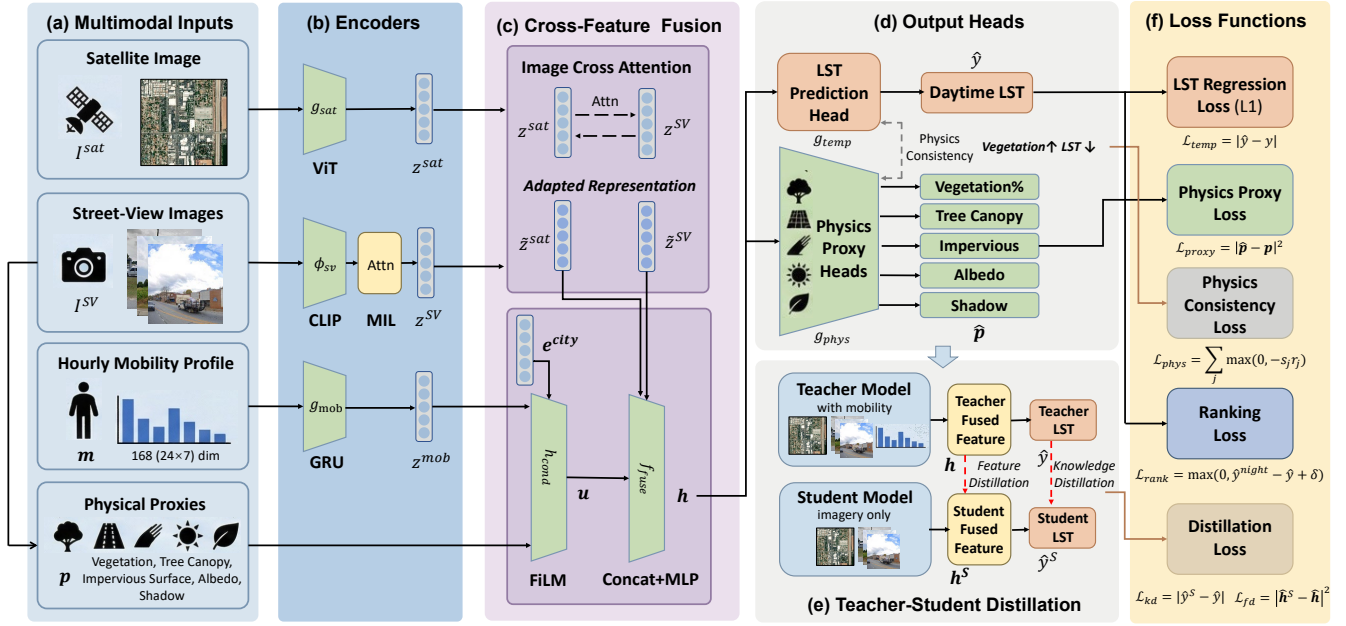


Figure 2: Overview of the proposed AESPA model.

3.3 Multimodal Encoders

Given the three input modalities, our model first maps each of them into a compact tract-level representation. A Vision Transformer-based [6] satellite encoder $g_{\text{sat}}(\cdot)$ maps the high-resolution satellite patch I_i^{sat} to a d_h -dimensional embedding $z_i^{\text{sat}} = g_{\text{sat}}(I_i^{\text{sat}}) \in \mathbb{R}^{d_h}$, summarizing large-scale land cover and morphological patterns around each tract.

For the set of street-view images $I_i^{\text{sv}} = \{I_{i,1}^{\text{sv}}, \dots, I_{i,K}^{\text{sv}}\}$, we first encode each view with a CLIP visual transformer encoder [32] $\phi_{\text{sv}}(\cdot)$ and then aggregate them with attention-based multiple instance learning (MIL) [13] to handle a variable number of views: $\alpha_{i,k} = \frac{\exp(\mathbf{w}^T \tanh(\mathbf{V}\phi_{\text{sv}}(I_{i,k}^{\text{sv}})))}{\sum_{k'} \exp(\mathbf{w}^T \tanh(\mathbf{V}\phi_{\text{sv}}(I_{i,k'}^{\text{sv}})))}$, $z_i^{\text{sv}} = \sum_k \alpha_{i,k} \phi_{\text{sv}}(I_{i,k}^{\text{sv}})$, yielding a single tract-level embedding that emphasizes thermally relevant street scenes.

For the teacher model, the 168-hour mobility sequence $\mathbf{m}_i$ is normalized and then passed through a 1-D convolution and a gated recurrent unit: $z_i^{\text{mob}} = g_{\text{mob}}(\mathbf{m}_i) \in \mathbb{R}^{d_m}$, capturing diurnal and weekly activity rhythms in tract i . The student model shares the same satellite and street-view encoders but omits g_{mob} , enabling deployment in settings without mobility data.

3.4 Street-View Physical Proxies

Beyond the learned street-view embedding, we derive five simple, physically motivated scalar proxies from the same set of street-view images. These quantities approximate well-known determinants of urban heat: vegetation and tree canopy are generally associated with lower land-surface temperatures [8, 46], whereas impervious, low-albedo, and unshaded surfaces tend to increase surface and air temperature and strengthen urban heat islands [37, 43]. For each tract i and view $I_{i,k}^{\text{sv}}$, we work in RGB space, denote the

luminance by $G = 0.299r + 0.587g + 0.114b$, and build five per-image proxies $\mathbf{p}_{i,k} \in \mathbb{R}^5$:

(i) *Vegetation*. We approximate a vegetation index by contrasting green and red channels $v_{i,k} = (g - r)/(g + r + 10^{-6})$ and take its image-wise mean [28].

(ii) *Tree canopy*. We combine a “green-dominant” mask $\{g > 100, g > r, g > b\}$ with an HSV-based green mask obtained from standard RGB-HSV conversion (hue in a green range, sufficient saturation and brightness). The proxy is the average of the two pixel fractions, clipped to $[0, 1]$ [25].

(iii) *Impervious surface*. We detect bright, low-saturation pixels $\{G > 150, S < 0.2\}$ as a visual proxy, and blend this fraction with an NDBI (Normalized Difference Built-up Index)-like term $(r - g)/(r + g + 10^{-6})$ using fixed weights (0.6 and 0.4) and clipping the result to $[0, 1]$.

(iv) *Albedo*. We use the mean luminance normalized to $[0, 1]$, i.e., $\alpha_{i,k} = \bar{G}/255$, clipped to $[0.1, 0.9]$ as proxies of albedo.

(v) *Shadow*. We take the fraction of very dark pixels $\{G < 50\}$ and clip it to $[0, 0.5]$.

For each tract, we aggregate over all views, $\mathbf{p}_i = \frac{1}{K} \sum_{k=1}^K \mathbf{p}_{i,k} \in \mathbb{R}^5$, obtaining an interpretable low-dimensional summary of the street-level physical context. These proxies are pre-computed from street-view images and later used both as conditioning signals and as targets of lightweight prediction heads in our model.

3.5 Cross-Feature Fusion and Prediction Heads

To capture interactions between modalities, our model fuses the encodings $z_i^{\text{sat}}, z_i^{\text{sv}}$, the proxy vector $\mathbf{p}_i$, and, for the teacher, the mobility embedding z_i^{mob} . We first apply a vector-level cross-attention module to let the satellite and street-view embeddings attend to

each other, producing $\tilde{\mathbf{z}}_i^{\text{sat}}$ and $\tilde{\mathbf{z}}_i^{\text{sv}}$. This step allows each view to adapt its representation in light of the other.

We then form a simple conditioning vector

$$\mathbf{u}_i = [\mathbf{e}_{\text{city}(i)} \parallel \mathbf{p}_i \parallel \mathbf{z}_i^{\text{mob}}],$$

where $\mathbf{e}_{\text{city}(i)}$ is a learnable city embedding and the mobility term is omitted for the student. A FiLM-style conditioner [31] $h_{\text{cond}}(\cdot)$ maps $\mathbf{u}_i$ to modulation parameters that rescale and shift the attended features, and a shallow multilayer perceptron $f_{\text{fuse}}(\cdot)$ produces the fused tract representation

$$\mathbf{h}_i = f_{\text{fuse}}(\tilde{\mathbf{z}}_i^{\text{sat}}, \tilde{\mathbf{z}}_i^{\text{sv}}, h_{\text{cond}}(\mathbf{u}_i)) \in \mathbb{R}^{d_h}.$$

On top of $\mathbf{h}_i$ we attach two types of lightweight heads. A temperature regression head $\hat{y}_i = g_{\text{temp}}(\mathbf{h}_i)$ produces the predicted daytime temperature. In parallel, a small linear–nonlinear head $\hat{\mathbf{p}}_i = g_{\text{phys}}(\mathbf{h}_i) \in \mathbb{R}^5$ recovers an estimate of the five physical proxies from the same representation. While only $\hat{y}_i$ is used at inference time, $\hat{\mathbf{p}}_i$ is used during training to keep $\mathbf{h}_i$ aligned with physically meaningful variations, which we exploit in the physics-aware regularization described next.

3.6 Physics-Aware Regularization

Our primary objective is to predict tract-level daytime temperature, but we exploit basic physical knowledge about urban heat to regularize the mapping from $\mathbf{h}_i$ to $\hat{y}_i$. The proxy predictions $\hat{\mathbf{p}}_i$ allow us to encode simple sign-level relations between urban form and temperature without imposing a rigid parametric model.

Proxy–temperature consistency. We focus on qualitative trends supported by urban climate studies: higher vegetation and tree canopy are associated with lower temperature, whereas higher imperviousness and stronger direct illumination are associated with higher temperature. Let $\hat{p}_{i,j}$ denote the j -th component of $\hat{\mathbf{p}}_i$ and $s_j \in \{-1, +1\}$ encode the expected sign of its correlation with temperature (e.g., $s_{\text{veg}} = -1$, $s_{\text{imp}} = +1$). For a mini-batch $\mathcal{B}$, we compute the empirical Pearson correlation $r_j = \text{corr}(\{\hat{p}_{i,j}\}_{i \in \mathcal{B}}, \{\hat{y}_i\}_{i \in \mathcal{B}})$ and penalize violations:

$$\mathcal{L}_{\text{phys}} = \sum_j \max(0, -s_j r_j).$$

This softly encourages the learned mapping to respect known monotone trends while leaving the exact functional form data-driven.

Day–night ordering. When nighttime temperature labels y_i^{night} are available, we attach an auxiliary head $g_{\text{night}}(\mathbf{h}_i)$ that predicts $\hat{y}_i^{\text{night}}$ and enforce a simple ordering constraint reflecting diurnal cycles. Specifically, we require daytime temperature to be no lower than nighttime temperature up to a small margin $\delta > 0$:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \max(0, \hat{y}_i^{\text{night}} - \hat{y}_i + \delta).$$

This auxiliary loss shapes the representation without changing the prediction target.

3.7 Mobility-Aware Teacher–Student Training

We emphasize that the teacher model is not intended to be the primary predictor during inference. Instead, its role is to provide

mobility-informed structural inductive bias to the student model during training. Given that real-world mobility signals are often noisy, incomplete, and highly city-specific, direct reliance on them at inference time could compromise generalization. The student model, distilled from the teacher, focuses on capturing transferable behavioral patterns while avoiding overfitting to raw mobility data. The teacher and student share the same satellite and street-view encoders and fusion module, but only the teacher consumes the mobility embedding $\mathbf{z}_i^{\text{mob}}$. The teacher produces fused representations $\mathbf{h}_i$ and temperatures $\hat{y}_i$, and is trained with the regression and physics-aware terms described above, leveraging all modalities.

The student model receives only satellite and street-view inputs and the tract-level proxies $\mathbf{p}_i$, yielding a fused representation $\mathbf{h}_i^S$ and a temperature prediction $\hat{y}_i^S = g_{\text{temp}}^S(\mathbf{h}_i^S)$.

To transfer the benefits of mobility-aware training without requiring mobility at inference, we align the student with the teacher through prediction and feature distillation:

$$\mathcal{L}_{\text{kd}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} |\hat{y}_i^S - \hat{y}_i|, \quad \mathcal{L}_{\text{fd}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\mathbf{h}_i^S - \mathbf{h}_i\|_2^2.$$

During student training, these distillation terms are combined with the standard regression and regularization objectives to shape $\mathbf{h}_i^S$ and $\hat{y}_i^S$ towards the mobility-enhanced teacher, while the deployed model remains imagery-only.

3.8 Training Objective

We train the teacher and student in two stages. For a mini-batch $\mathcal{B}$, the teacher minimizes

$$\mathcal{L}_{\text{teacher}} = \mathcal{L}_{\text{temp}} + \lambda_{\text{proxy}} \mathcal{L}_{\text{proxy}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}$$

where $\mathcal{L}_{\text{temp}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} |\hat{y}_i - y_i|$ is the temperature regression loss and $\mathcal{L}_{\text{proxy}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2^2$ encourages accurate reconstruction of the proxies.

In the second stage, we freeze the teacher and optimize the student with

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{temp}}^S + \lambda_{\text{proxy}} \mathcal{L}_{\text{proxy}}^S + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{kd}} \mathcal{L}_{\text{kd}} + \lambda_{\text{fd}} \mathcal{L}_{\text{fd}},$$

with $\mathcal{L}_{\text{temp}}^S$ and $\mathcal{L}_{\text{proxy}}^S$ defined analogously.

4 Experiments and Results

In this section, we evaluate our proposed AESPA model to answer the following research questions:

- **RQ1:** Can AESPA outperform satellite-only and street-view-only baselines on tract-level daytime temperature prediction across diverse U.S. metropolitan areas?
- **RQ2:** How do different input modalities and physics-aware components (proxy heads, physics-consistency loss, and day–night ranking loss) contribute to the accuracy and robustness of AESPA?
- **RQ3:** Does the model improve cross-MSA generalization compared with purely imagery-based models?
- **RQ4:** Can AESPA reveal intra-urban heat disparities across neighborhoods with different socioeconomic characteristics?

Table 1: LST prediction results on 8 U.S. MSAs. Best results are in bold and the second best results are underlined.

MSA	Dallas		Washington		Miami		Boston		Seattle		Minneapolis		St. Louis		Pittsburgh	
Metrics	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ
ResNet	1.7710	0.5966	1.7013	0.4407	1.4232	0.4819	2.2181	0.8292	3.1445	0.3698	2.1117	0.6660	2.0136	0.7714	1.1821	0.7586
Tile2vec	4.6536	0.5492	1.9042	0.4742	5.0082	0.0056	3.0756	0.7829	1.8170	0.6024	2.6846	0.5938	3.8418	0.7851	1.8419	0.6963
UrbanHeat	8.5411	0.4859	6.6003	0.3924	1.5009	0.3966	5.6580	0.8374	6.5082	0.3833	3.1428	0.5626	2.4598	0.6619	3.1477	0.7032
Proxy + Regression	1.7935	-0.1969	1.9574	0.0491	1.8559	0.0628	2.8680	0.4181	2.2315	0.5244	2.3418	-0.0187	2.0517	-0.2260	2.35	0.1713
AESPA	0.8926	0.8499	<u>1.4112</u>	0.6398	1.1817	0.7688	1.1627	0.9049	1.5904	<u>0.7740</u>	<u>1.4763</u>	0.7173	1.3938	0.7800	1.5153	0.6584
w/o Satellite	0.8835	0.8459	1.5637	0.6135	1.1539	0.7578	1.1525	0.9029	2.2495	0.4887	1.5681	0.6159	1.5931	0.7546	1.5367	0.6645
w/o Street View	0.9226	0.8412	1.4486	0.6162	1.1167	0.7877	<u>1.0888</u>	0.9088	1.7926	0.7270	1.6100	0.6289	1.4124	0.7666	1.4746	0.6632
w/o Mobility	0.9840	0.8235	1.5346	0.6041	1.1428	0.7895	<u>1.1464</u>	<u>0.9134</u>	<u>1.6095</u>	0.7789	1.4508	<u>0.6928</u>	1.5398	0.7794	1.4877	0.6375
Satellite Only	0.9265	0.8358	1.5642	0.5546	1.1202	0.7728	1.0660	0.9146	1.9609	0.6904	1.5830	0.6840	1.4387	0.7895	1.6166	0.633
Street View Only	0.9246	0.8469	1.4947	0.6164	<u>1.0432</u>	<u>0.8023</u>	1.1310	0.8985	1.7819	0.7575	1.5301	0.6526	1.3499	0.7814	1.5164	0.6368
w/o $\mathcal{L}_{phys}$	0.8951	<u>0.8471</u>	1.4747	0.6147	1.1544	0.7544	1.1045	0.9087	1.6981	0.7478	1.6026	0.6729	1.3897	0.7945	1.4791	0.669
w/o $\mathcal{L}_{proxy}$	0.9627	0.8135	1.3726	0.6171	1.0837	0.8089	1.1771	0.8949	1.8172	0.7190	1.7175	0.6280	<u>1.3706</u>	<u>0.8046</u>	1.5032	0.6724
w/o $\mathcal{L}_{rank}$	0.9821	0.8083	1.4939	<u>0.6283</u>	1.1190	0.7849	1.1607	0.8892	1.8316	0.7306	1.5867	0.6587	1.4068	0.8104	1.4752	0.6674
w/o Distillation	0.9110	0.8402	1.4570	0.6061	1.1259	0.7536	1.1692	0.9018	1.9119	0.7226	1.6927	0.5633	1.4093	0.7867	<u>1.4446</u>	<u>0.7052</u>

4.1 Experimental Settings

Datasets. We evaluate our model using multi-modal urban datasets collected from 8 metropolitan statistical areas (MSAs) in the United States. These datasets integrate socioeconomic indicators, satellite and street-view imagery, human mobility patterns, and Land Surface Temperature (LST) records; the details of the datasets can be found in Appendix A.

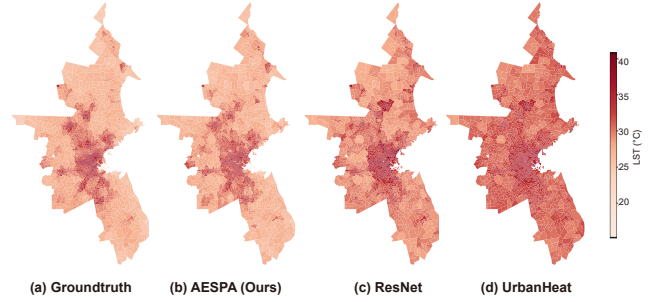
Baseline Models. We compare AESPA against following representative baselines: (i) **ResNet** [9] uses an ImageNet-pretrained ResNet-18 encoder applied separately to satellite and street-view images, followed by global average pooling and feature concatenation at the tract level. (ii) **Tile2Vec** [15] employs the same fusion architecture as the ResNet baseline, but initializes the ResNet-18 encoder with Tile2Vec-pretrained weights. (iii) **UrbanHeat** [36] follows a satellite-only design based on multi-scale context. It extracts satellite tiles centered at the tract centroid with approximate ranges of 250 m and 500 m, and apply a cross-attention module to fuse the two scales before regressing LST. (iv) **Proxy+Regression** uses the street-view-derived physical proxies aggregated at the tract level as input to a linear regression model.

Training and Evaluation Metrics. Training details can be found in Appendix B. We evaluate tract-level temperature prediction under two protocols: (i) *within-MSA* prediction, where models are trained and tested on tracts from the same MSA, and (ii) *cross-MSA* prediction, where models are trained on one MSA and directly evaluated on a different held-out MSA.

For both settings, we report two standard metrics on the corresponding test sets: First, the *Mean Absolute Error* (MAE) measures the average absolute difference between predicted and observed daytime LST values, with lower MAE indicating higher accuracy. Second, we compute the *Pearson correlation coefficient* ρ between predicted and observed tract-level temperatures, which quantifies how well the model preserves the spatial ranking of hotter and cooler tracts; higher ρ indicates better agreement.

4.2 Performance Evaluation

Overall Performance (RQ1). Table 1 reports within-MSA daytime LST prediction results across the eight U.S. MSAs. AESPA achieves the lowest MAE in 7 out of 8 MSAs and the highest or

**Figure 3: Groundtruth (a) and predicted LST of AESPA (b), UrbanHeat (c) and ResNet (d) in Boston MSA.**

second-highest Pearson correlation ρ in all cases. On average, AESPA reduces MAE from 1.95°C for the strongest baseline (ResNet) to 1.33°C (a relative reduction of 32%), while increasing the mean correlation from 0.61 to 0.76. In individual cities, the gains are particularly pronounced in dense coastal MSAs such as Boston and Seattle, where AESPA lowers MAE by more than 1°C compared with ResNet and improves ρ by around 0.1–0.2. Only in Pittsburgh does ResNet obtain a slightly better performance, but AESPA remains competitive, indicating that the multimodal design is generally robust across diverse urban forms.

Figure 3 provides a visual comparison for the Boston MSA. AESPA’s predictions (b) closely reproduce both the cooler corridors and the localized hot spots in dense built-up areas visible in the ground-truth map (a), whereas UrbanHeat and ResNet (c-d) tend to over-smooth city-centre temperatures and miss cooler pockets embedded within otherwise warm regions. This agreement indicates that AESPA not only improves numerical accuracy but also better preserves fine-grained differences in heat levels across census tracts. **Ablation Studies (RQ2).** The lower part of Table 1 reports ablations that progressively remove AESPA’s key components. We first examine the role of multimodal fusion. Removing either satellite or street-view imagery (*w/o Satellite*, *w/o Street View*) increases the average MAE by about 6–10% and reduces the mean tract-level

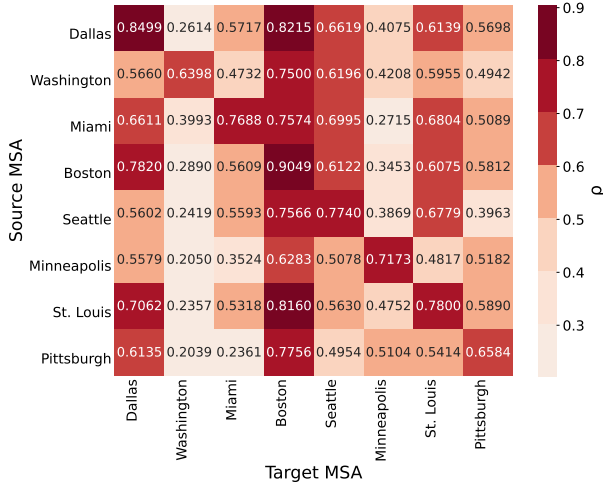


Figure 4: Comparison of performance (Pearson correlation ρ) evaluated on cross-MSA experiments of AESPA model.

correlation P by roughly 0.06. The degradation is especially pronounced in Seattle and Minneapolis, where MAE increases by up to 0.66°C and ρ drops by about 0.30 when one view is removed, indicating that both nadir-view and street-level cues are needed to capture complex microclimates in dense, heterogeneous MSAs. Training single-view variants (*Satellite Only*, *Street View Only*) also underperforms the full model on average, although *Street View Only* comes close to AESPA in some MSAs, suggesting that street-level morphology carries strong predictive power that is further refined by satellite context. When mobility profiles are removed (*w/o Mobility*), performance declines more modestly (around 3% higher MAE and slightly lower ρ), but the drop is consistent, confirming that mobility-informed supervision provides additional signal beyond static imagery.

We then investigate the physics-aware objectives. Disabling the physics-consistency, proxy, or day–night ranking losses (*w/o $\mathcal{L}_{phys}$* , *w/o $\mathcal{L}_{proxy}$* , *w/o $\mathcal{L}_{rank}$*) all lead to worse average performance, with MAE increasing by 2–6% and ρ decreasing by 0.01–0.02. The largest degradations occur for *w/o $\mathcal{L}_{proxy}$* and *w/o $\mathcal{L}_{rank}$* , particularly in Seattle and Minneapolis, indicating that enforcing monotonic relationships with street-view proxies and the day–night ordering helps stabilize learning in climates with strong diurnal contrast.

Finally, removing the teacher–student distillation (*w/o Distillation*) results in an average MAE increase of about 5% and a notable drop in ρ (e.g., from 0.72 to 0.56 in Minneapolis), showing that guidance from the mobility-augmented teacher is important even for within-MSA prediction. Overall, these ablations support our design choices: fusing satellite and street-view imagery, incorporating lightweight physics-aware losses, and distilling from a mobility-informed teacher each make complementary contributions to the accuracy and robustness of AESPA. This suggests that the distillation process enables the student to internalize essential behavioral structures while remaining resilient to the noise and city-specific biases inherent in raw mobility data, ultimately leading to superior generalization.

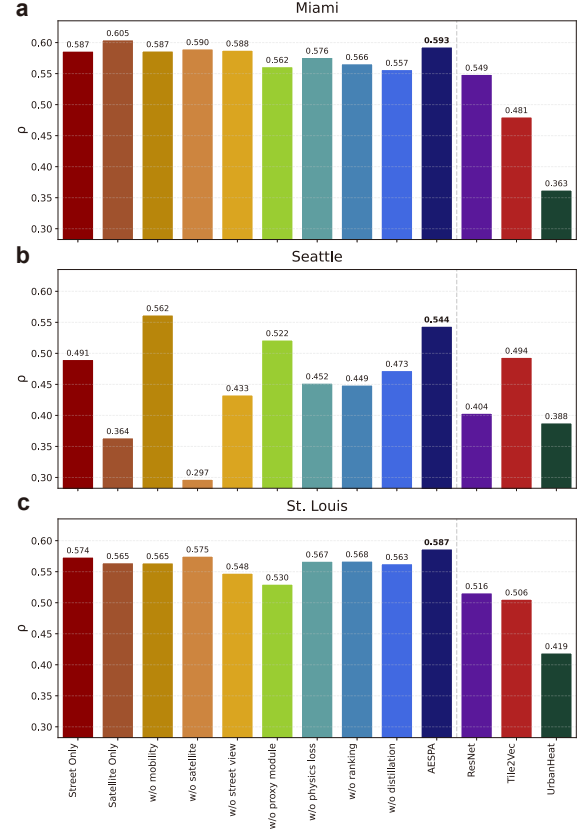


Figure 5: Average ρ on cross-MSA experiments with Miami (a), Seattle (b), and St. Louis (c) as source MSA.

Cross-MSA Prediction (RQ3). Figure 4 summarizes the cross-MSA performance of AESPA (student). As expected, diagonal entries (train and test on the same MSA) are highest, but most off-diagonal cells still reach $\rho \approx 0.5$ –0.7, indicating that a model trained in one MSA can recover much of the relative temperature ordering across census tracts in other MSAs without target labels. Transfers between climatically similar MSAs (e.g., Miami→Dallas, Boston→St. Louis) achieve the strongest correlations, whereas transfers into Washington and Minneapolis are consistently weaker, suggesting that these MSAs exhibit more distinctive climatic conditions.

Figure 5 compares AESPA with ablated variants and baselines by averaging ρ over all targets for three source MSAs. Across Miami, Seattle, and St. Louis, AESPA attains the highest average correlation and improves over ResNet and UrbanHeat by about 0.05–0.10. Single-view models (*Street Only*, *Satellite Only*) and variants without physics-aware losses or the proxy module perform noticeably worse, and removing distillation further reduces ρ for all three sources. These results show that multimodal fusion, lightweight physics constraints, and teacher–student distillation all contribute to AESPA’s robustness when transferring to unseen MSAs.

Case Study: Socioeconomic Heat Gradients in Dallas (RQ4). To examine whether AESPA can reveal intra-urban heat disparities across socioeconomic groups, we conduct a case study for

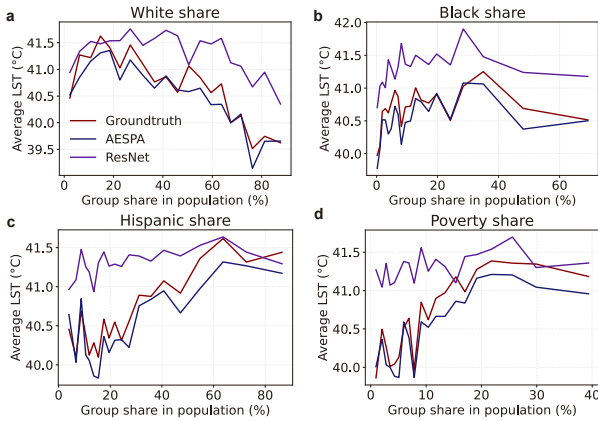


Figure 6: Average LST as a function of Dallas MSA’s tract-level socioeconomic features, comparing ground truth, AESPA, and ResNet.

the Dallas MSA. For each census tract, we compute the share of White, Black, and Hispanic residents and the share of people living in poverty, then group tracts into bins by group share and plot the average daytime LST for ground truth, AESPA, and the best-performing baseline in Figure 6. The ground-truth curves show clear socioeconomic gradients: tracts with lower White share and higher Hispanic or poverty share tend to be hotter, while LST peaks at intermediate Black shares. AESPA closely follows these patterns in both level and slope, whereas ResNet systematically overestimates temperatures and attenuates several gradients, particularly at high White shares and low poverty shares. This suggests that AESPA not only improves overall numerical accuracy but also better preserves the underlying relationships between heat exposure and neighborhood composition, enabling a more faithful characterization of intra-urban heat disparities.

5 Discussions

Our findings show that web-based multimodal data can substantially enhance tract-level estimation of urban land surface temperature. By combining satellite imagery, street-view panoramas, mobility-derived activity profiles, and physically motivated proxies, AESPA consistently outperforms satellite-only and proxy-only baselines in both within-MSA and cross-MSA settings. Performance variations across cities are expected due to differences in urban morphology, data availability, and imaging conditions. In particular, methods relying on a single modality may be more sensitive to city-specific characteristics. The framework fits naturally into the vision of using web resources to support sustainable and equitable urban services, complementing emerging open workflows and dashboards for mapping urban heat and equity risks [7, 33].

These improvements carry direct implications for climate equity and the Sustainable Development Goals, particularly SDG 10 and SDG 11 [29]. A growing body of work has documented that lower-income and racially marginalized neighborhoods are systematically hotter than wealthier, predominantly White areas in U.S. cities [3, 11, 45]. AESPA’s tract-level maps recover similar spatial disparities

from web-based data alone, and its proxy heads attribute elevated temperatures to combinations of reduced vegetation and canopy, higher impervious surfaces, and lower shading, consistent with established urban-climate mechanisms. Coupled with web-based visualization tools, these outputs could underpin interactive heat-risk dashboards that help city agencies, NGOs, and communities identify where heat burdens and adaptation deficits are concentrated and which levers (e.g., urban greening, cool roofs, shading) are most relevant.

Methodologically, AESPA illustrates a pragmatic recipe for physics-aware multimodal learning in environmental applications. Recent work on physics-informed machine learning for weather and climate has shown that embedding domain constraints into neural networks can improve robustness and interpretability [4, 16, 41, 51]. Instead of relying on detailed forcings or high-resolution 3D simulators, we encode a small set of well-established physical relationships (e.g., monotonic links between vegetation proxies and temperature) and a simple day–night ranking constraint as soft losses within a multimodal encoder. The teacher–student design further shows how richer modalities such as mobility, increasingly used to characterize thermal exposure and mobility–heat interactions [26, 44], can be exploited during training and then distilled into an imagery-only student, improving cross-city generalization while keeping deployment lightweight and compatible with data-poor environments.

This work has limitations. Empirically, we focus on eight U.S. metropolitan areas with high data coverage; however, it remains to be seen how the framework generalizes to regions with sparse web data, distinct urban forms, or extreme climates. Furthermore, the physics-aware components in AESPA are designed as directionally consistent regularization rather than detailed physical simulations. While this maintains computational scalability, these priors remain relatively coarse and do not explicitly account for temporal dynamics, extreme heatwaves, or long-term adaptation trends. Future work could incorporate additional modalities – such as nighttime lights or semantic 3D information – and integrate richer physical constraints while preserving model efficiency. Finally, embedding AESPA into interactive web platforms that couple heat estimates with health, energy, or mobility indicators would further align with the Web4Good agenda. By transforming web-based imagery and mobility into transparent, deployable tools, such extensions can better support climate equity and heat risk mitigation.

Ethical Use of Data. Protecting individual privacy is a paramount consideration throughout our study. The urban imagery data employed for our model training originates from publicly available sources with privacy-protective licenses². These images are coarse-grained in resolution, preventing the identification of individuals. The mobility patterns, socioeconomic indicators, and LST are all aggregated at the tract level, ensuring no individual targeting. Consequently, our research complies with ethical data usage standards.

Acknowledgments

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²<https://trust.arcgis.com/en/privacy/privacy-tab-intro.htm>

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A Dataset Details

Table 2: Basic statistics of 8 metropolitan statistical areas (MSAs) in the U.S.

MSA	Average LST (°C)	#Census Tracts
Dallas	40.7	1,312
Washington	33.57	1,359
Miami	37.7	1,216
Boston	31.54	1,003
Seattle	31.82	718
Minneapolis	31.29	785
St. Louis	34.88	615
Pittsburgh	30.52	711

Study Areas. We focus on 8 metropolitan statistical areas (MSAs) in the U.S. and take census tract as our basic spatial unit. The MSAs are Dallas-Fort Worth-Arlington, TX Metro Area (shortened as “Dallas”), Washington-Arlington-Alexandria, DC-VA-MD-WV Metro Area (“Washington”), Miami-Fort Lauderdale-Pompano Beach, FL Metro Area (“Miami”), Boston-Cambridge-Newton, MA-NH Metro Area (“Boston”), Seattle-Tacoma-Bellevue, WA Metro Area (“Seattle”), Minneapolis-St. Paul-Bloomington, MN-WI Metro Area (“Minneapolis”), St. Louis, MO-IL Metro Area (“St. Louis”), Pittsburgh, PA Metro Area (“Pittsburgh”). Socioeconomic indicators for each census tract are gathered from the 2019 American Community Survey (ACS) 5-year estimates³.

Urban Imagery. Urban imagery is collected from web-based mapping platforms. For every census tract, we obtain one nadir-view satellite tile covering the tract and its immediate surroundings from Esri⁴ and resize it to 256×256 pixels for model input. In addition, we sample up to 40 street-view panoramas along public roads within each tract. These panoramas are downloaded via the Google Street View API⁵ and pre-processed (cropping, resizing, and normalization) before being encoded into tract-level visual representations.

Mobility Data. Human mobility data are derived from the SafeGraph Weekly Patterns⁶ product for August to October 2019. For each census tract, we aggregate the hourly visitor counts of all points of interest (POIs) located within the tract into a 7×24 histogram. Summing over POIs and normalizing the histogram by its total count yields a 168-dimensional mobility profile $\mathbf{m}$, which characterizes the distribution of visits to that tract over a week on average.

LST Data. Tract-level LST is sourced from the United States Surface Urban Heat Island database [2]. We extract summer daytime LST as the prediction target; the mean LST values for the eight MSAs are detailed in Table 2. While different modalities operate at varying native temporal resolutions, all dynamic signals are temporally aggregated to match the prediction horizon. Consequently, the model learns integrated heat exposure patterns within a consistent temporal window rather than modeling fine-grained, instantaneous dynamics.

B Implementation Details

All models are trained for up to 30 epochs on a single NVIDIA A100 GPU. We first train the teacher model and then distill its knowledge into the student model. Optimization is performed with the AdamW optimizer [24], using a learning rate of 1×10^{-4} and a weight decay of 0.05, together with early stopping based on validation performance. For the teacher stage, we set the loss weights to $\lambda_{\text{phys}} = 0.05$, $\lambda_{\text{proxy}} = 0.0$, and $\lambda_{\text{rank}} = 0.1$. For the student stage, we use $\lambda_{\text{phys}} = 0.2$, $\lambda_{\text{proxy}} = 0.3$, $\lambda_{\text{rank}} = 0.1$, $\lambda_{\text{kd}} = 0.1$, and $\lambda_{\text{fd}} = 0.05$. Within each MSA, data are split into 60%, 20%, and 20% of tracts for training, validation, and testing, respectively. We repeat all experiments with five random splits and report the average performance on test sets.

³<https://www.census.gov/programs-surveys/acs>

⁴<https://learn.arcgis.com/en/projects/download-imagery-from-an-online-database/>

⁵<https://developers.google.com/maps/documentation/streetview/overview>

⁶<https://docs.deweydata.io/docs/advan-research-weekly-patterns>